



OPEN CV 5 · AGENTIC VISION

Evidence before explanation.

A reproducible OpenCV 5 pipeline that turns visual changes into sealed evidence, then lets a bounded agent rerun, seal, complete, or request human approval.

31 / 31

local tests passed

2 TP · 2 TN

seeded smoke evaluation

SHA-256

canonical evidence receipt

STATUS

LOCAL + CONTAINER VERIFIED · AWS LIVE RUN PENDING



ONE EVIDENCE PATH

From video bytes to bounded action.

Observation, interpretation, and action are separate stages. The agent never substitutes prose for visual evidence.

**CORE INVARIANT**

A later explanation may cite only known frame IDs and must preserve the action selected by the OpenCV evidence policy. Unknown citations or unapproved recommendations fail closed.

SUBSTANTIVE VISION PIPELINE

Decode. Compare. Localize. Seal.

OpenCV 5.0.0.93 is pinned and used across decoding, comparison, localization, and review rendering.

1

Decode

VideoCapture reads frames and frame rate.

2

Preview

Grayscale + INTER_AREA downsampling.

3

Compare

Normalized absdiff produces a change score.

4

Localize

Threshold, morphology-close, contours, rectangles.

5

Stabilize

Hysteresis suppresses continuous-motion floods.

6

Review

Separate overlay; source frames stay unchanged.

ADVERSARIAL FIX

One public 125-frame clip initially produced 56 cards. Hysteresis reduced that to one without disabling re-arming after a quiet frame.



EVIDENCE CHANGES THE ACTION

Four actions. No invented fifth.

The planner is deterministic and fail-closed. Material transitions remain human-owned.

OpenCV evidence	Permitted action	Reason
No card over threshold	complete_no_material_change	No visual transition was extracted.
Card has no region	rerun_region_extraction_then_review	Do not summarize unlocalized evidence.
Localized card	seal_evidence_for_agent_summary	A later summary may cite the card.
Score ≥ 0.25	request_human_approval	A material transition requires a person.

FAIL-CLOSED CHECK

The explanation module rejects unknown frame citations and recommendations that do not exactly match the bounded plan.



SEEDED ENGINEERING SMOKE TEST

Small test. Explicit limits.

Four deterministic scenarios assert both event detection and the selected policy action.

Scenario	Expected event	Detected	Action
No change	No	No (TN)	complete
Localized change	Yes	Yes (TP)	seal
Full-frame change	Yes	Yes (TP)	human approval
Low-amplitude noise	No	No (TN)	complete

2 TP - 2 TN

0 false positives - 0 false negatives

4 / 4

expected actions matched

Public-sample checks

Frames	Events	Action
125	1	seal evidence
15	1	seal evidence
54	2	seal evidence

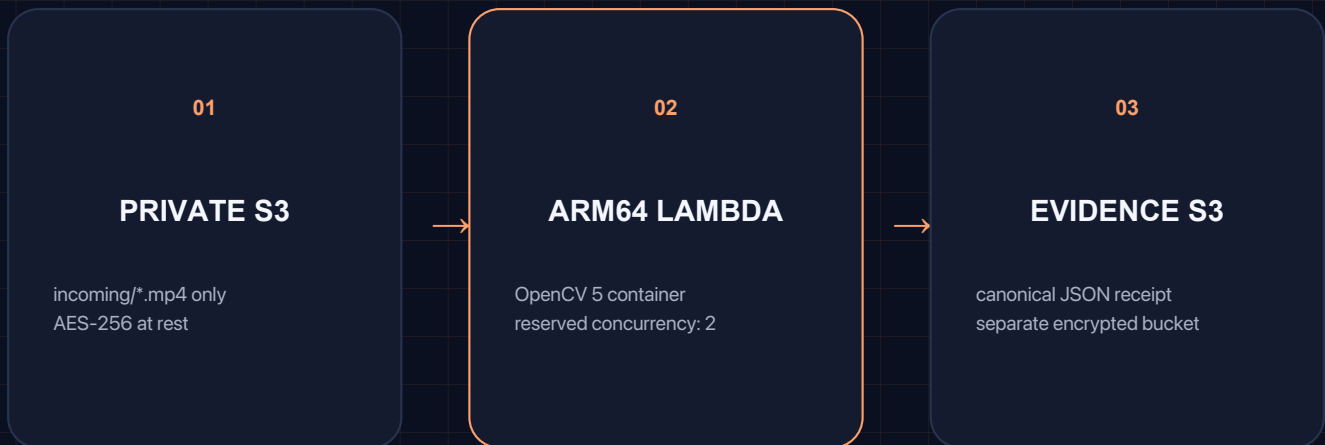
Limit: these are engineering gates and public-format checks, not a held-out real-world benchmark or production accuracy claim.



BOUNDED CLOUD DELIVERY

Private input. Encrypted evidence.

The SAM template and Lambda-compatible arm64 container are locally verified. A live AWS deployment remains pending.

**RESPONSIBLE OPERATING BOUNDARY**

- Reject multiple S3 records and output-recursion events.
- Default object retention: 7 days; configurable 1–30 days.
- CloudWatch log retention: 14 days.
- No public buckets; report location returned by Lambda invocation.
- Right-cleared demo media only; deployment claims require live proof.



RESPONSIBLE AGENTIC VISION

See the change. Verify the action.

Evidence Locker is designed for inspection and review workflows where source frames, changed regions, and the reason for a next action must remain auditable.

Known limits

- Change detection is not object recognition, identity verification, intent inference, or a safety-critical anomaly detector.
- Thresholds are content-dependent; subtle changes can be missed and camera motion or illumination can trigger events.
- The four-case suite is deliberately smoke-sized. Production use needs a right-cleared, held-out evaluation set.
- Hashes prove byte-level integrity, not the real-world meaning of a scene.
- Material transitions route to human approval; the agent cannot authorize irreversible actions.

REPRODUCE

```
python3 -m venv .venv && .venv/bin/pip install -r requirements.txt

.venv/bin/python -m unittest -v

.venv/bin/python evaluate_synthetic.py

docker build --platform linux/arm64 -t evidence-locker .
```

SOURCE

github.com/tzh476/opencv-evidence-locker

NO AWARD OR PAYMENT CLAIMED · LIVE AWS EVIDENCE WILL BE ADDED ONLY AFTER VERIFICATION